

Ana Lorena Ortiz Loyola

Frankfurt am Main, Germany

☎ +49 176 62145797 • ✉ a01750283@tec.mx • [LinkedIn](#)

Data science student (Tecnológico de Monterrey, expected 2027) with hands-on machine learning experience (Scikit-Learn, Keras, TensorFlow) and object-oriented programming in Python and Java, combined with practical financial-services experience through a data engineering internship at Deutsche Bundesbank (AnaCredit). Strong analytical foundation from independent research in cryptography and predictive modeling, with a growing interest in applying AI to capital-markets business.

Core Competencies

- **Machine Learning & AI:** Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means, predictive modeling
- **Object-Oriented Programming:** Python, Java, C++
- **Financial Services Experience:** Data quality verification, migration testing, regulatory reporting (Deutsche Bundesbank AnaCredit)
- **Analytical Rigor:** Statistics, numerical optimization, cryptographic complexity analysis
- **Data & Databases:** SQL, SQLite, Pandas, NumPy
- **Languages:** German (C1), English (C1)

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reported banking data

Projects

- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of national health data; built and validated predictive models (Random Forest, Neural Networks) and K-Means clustering
- **Personal Project (GitHub), built with Claude Code**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a fully auditable data-cleaning log
- **Tecnológico de Monterrey**
Air Quality Analysis Mexico City – SVM Classification 04/2023–06/2023
 - Built SVM models (linear and RBF kernels) to classify critical ozone levels from citywide NO_x/O₃ data

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: machine learning & AI, data analysis & statistics, numerical optimization.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.